MATH 3330 Homework 6

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Question 1

Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on X. If $\mathcal{T} \subset \mathcal{T}'$, what does connectedness of X in one imply about connectedness in the other? What does the compactness of X under one of these topologies imply about compactness under the other?

If $\mathcal{T} \subset \mathcal{T}'$ and $\mathcal{T}'$ is connected, then $\mathcal{T}$ must also be connected because it is a coarser topology.

On the other hand, if $\mathcal{T} \subset \mathcal{T}'$ and $\mathcal{T}$ is compact, then $\mathcal{T}'$ must also be compact. This is because $\mathcal{T}$ has more sets and thus more possible open covers.

Question 2

Show that if X is an infinite set, it is connected in the finite complement topology (sets are open if their complement is finite).

Proof: Let X be an infinite set with the cofinite topology, meaning sets in X are open if their complement is finite. We must show that X is connected, so there should not exist nonempty open sets U and V such that $U \cap V = \varnothing$ and $U \cup V = X$. Assume for the sake of contradiction that X is disconnected, so sets U and V should exist.

If $U \subseteq X$ is nonempty, then $X \setminus U$ is finite because U is infinite. Similarly if $V \subseteq X$ is nonempty, then $X \setminus V$ is finite because V is finite. Since $U \cup V = X$, $X \setminus U = V$ and $X \setminus V = U$, however the union of two finite sets is finite, contradicting the fact that X is infinite.

Therefore, sets U and V must not exist and X must be connected.

Question 3

Determine whether or not $\mathbb{R}^\omega$ is connected in the uniform topology.

$\mathbb{R}^\omega$ is disconnected because we can find two sets $A = \{(x_n) \in \mathbb{R}^\omega \mid (x_n) \text{ is bounded}\}$ and $B = \{(x_n) \in \mathbb{R}^\omega \mid (x_n) \text{ is unbounded}\}$. $A \cup B = \mathbb{R}^\omega$ and $A \cap B = \emptyset$. Both sets are nonempty because we can find an element in each A and B: $(0, 0, 0, \ldots) \in A$ and $(0, 1, 2, 3, \ldots) \in B$.

Since both A and B are open, disjoint, and partition $\mathbb{R}^\omega$, these sets form a separation and thus $\mathbb{R}^\omega$ is not connected.

Question 4

Show that no two of $(0,1)$, $(0,1]$ and $[0,1]$ are homeomorphic. (Hint: What happens if you remove a point from each of these spaces?)

Proof: Consider the sets $A = (0,1)$, $B = (0,1]$, and $C = [0,1]$. Let $f : A \to B$, $g : B \to C$, and $h : C \to A$ be functions. We will show that no two of A, B, or C are homeomorphic by showing that f, g, and h are not continuous bijective functions with continuous inverses.

Consider $f : (0,1) \to (0,1]$.
Let $p = f^{-1}(1) \in (0,1)$. If we restrict f by removing the point p, then f becomes $f|_{(0,1)\setminus\{p\}} : (0,1) \setminus \{p\} \to (0,1] \setminus \{1\}$. Then the domain $(0,1) \setminus \{p\} = (0,p) \cup (p,1)$ and must be disconnected, however $(0,1] \setminus \{1\} = (0,1)$ which is connected. f can't be a homeomorphism because it does not preserve connectedness.

Consider $g : (0,1] \to [0,1]$.
Let $p = g^{-1}(0) \in (0,1]$. If we restrict g by removing the point p, then g becomes $f|_{(0,1]\setminus\{p\}} : (0,1] \setminus \{p\} \to [0,1] \setminus \{0\}$. Then the domain $(0,1] \setminus \{p\} = (0,p) \cup (p,1]$ and must be disconnected, however $[0,1] \setminus \{0\} = (0,1)$ which is connected. g can't be a homeomorphism because it does not preserve connectedness.

Consider $h : [0,1] \to (0,1)$
Since $[0,1]$ is compact in $\mathbb{R}$ and $(0,1)$ is not compact in $\mathbb{R}$ and homeomorphisms must preserve compactness, $h : [0,1] \to (0,1)$ must not be a homeomorphism.

Therefore, none of f, g, h are homeomorphisms.

Question 5

Let $f : S^1 \to \mathbb{R}$ be a continuous map. Show there exists a point x of S^1 such that $f(x) = f(-x)$.

Proof: Let $f : S^1 \to \mathbb{R}$ be a continuous map. We must show that there exists a point $x \in S^1$ such that $f(x) = f(-x)$. Define a new continuous function $g : S^1 \to \mathbb{R}$ as $g(x) = f(x) - f(-x)$. We must show that $g(x) = 0$.

Fix a point $y \in S^1$. We have

$$g(-y) = f(-y) - f(y) = -(f(y) - f(-y) = -g(y)$$

If $g(y) = 0$ for the fixed point y, trivially we are done. If $g(y) \neq 0$, then $g(y)$ and

$g(-y) = -g(y)$ have opposite signs. Since g is continuous (because it is a combination of continuous functions), by the intermediate value theorem, there there must exist some point $y' \in S^1$ such that $g(y') = 0$. Therefore, $f(x) = f(-x)$.

> Question 6
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> Show that if X is a well ordered set, then $X \times [0,1)$ in the dictionary order is a linear continuum.

Proof: Let X be a well ordered set and $A = X \times [0,1)$ have the dictionary ordering. We must show that A is a linear continuum. For A to be a linear continuum, A must have the least upper bound property and for each $x, y \in A$, if $x < y$, there must exist a $z \in A$ such that $x < z < y$.

Since A has the dictionary ordering, $(x_1, y_1) < (x_2, y_2)$ if $x_1 < x_2$ or ($x_1 = x_2$ and $y_1 < y_2$). Take any two points $(x_1, y_1) < (x_2, y_2)$ in A There are two cases:

- If $x_1 = x_2$, then $y_1 < y_2$ and since $[0,1)$ is dense, there must always exist some z such that $y_1 < z < y_2$. So then $(x_1, y_1) < (x_1, z) < (x_2, y_2)$.
- If $x_1 < x_2$, let $z = \frac{y_1+1}{2} \in (y_1, 1)$. Then $(x_1, y_1) < (x_1, z)$ since $x_1 = x_1$ and $y_1 < z$, and $(x_1, z) < (x_2, y_2)$ since $x_1 < x_2$. Thus $(x_1, y_1) < (x_1, z) < (x_2, y_2)$.
  Therefore we can always find a point in A that's between two other points in A.

Now let $B \subseteq A$ be nonempty and bounded above. We want to show $\sup(B)$ exists in A (A has the LUB property). Let $B_X = \{x \in X \mid (x, y) \in B \text{ for some } y\}$ be the projection of B onto X. Since B is bounded above, B_X is bounded above in X. Because X is well-ordered, B_X has a least upper bound $x_0 \in X$.

Now let $T = \{y \in [0,1) \mid (x_0, y) \in B\}$.

- If T is nonempty, then $T \subseteq [0,1)$ is bounded above, so it has a least upper bound $y_0 = \sup T$ in $[0,1)$. Then $(x_0, y_0) = \sup B$.
- If T is empty, then all elements of B have first coordinate strictly less than x_0, so $(x_0, 0)$ is an upper bound for B. Thus $(x_0, 0) = \sup B$.
  In both cases $\sup B$ exists in A, so A has the least upper bound property.

Therefore, X is a linear continuum because it has the LUB property and always has a third point between two others in the set.

> Question 7
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> Show that the finite union of compact spaces in compact.

Proof: Let A and B be compact spaces. Let $\{U_\alpha\}$ be an open cover for $A \cup B$. $\{U_\alpha\}$ covers A, and since A is compact, there exist finitely many U_{α_i} such that $A \subseteq U_{\alpha_1} \cup \ldots \cup U_{\alpha_n}$. Similarly, $\{U_\alpha\}$ covers B, and since B is compact, there exist finitely many U_{β_i} such that $B \subseteq U_{\beta_1} \cup \ldots \cup U_{\beta_n}$. Then the collection $\{U_{\alpha_1} \cup \ldots \cup U_{\alpha_n}, U_{\beta_1} \cup \ldots \cup U_{\beta_n}\}$ forms a finite open cover of $A \cup B$, so $A \cup B$ is compact. (The union of any two sets is compact)

Then by induction, since the union of any two sets A and B is a compact set (call it C), then $C \cup D$ for a compact set D is also compact because it is a union of two compact sets. Therefore the union of any finite number of compact sets is compact. :LiBookCheck:

> Question 8
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> Show that if $f : X \to Y$ is continuous, where X is compact and Y is Hausdorff, then f is a closed map (i.e. f maps closed sets to closed sets.)

Proof: Let $f : X \to Y$ be a continuous map where X is compact and Y is Hausdorff. We must show that f is a closed map that maps closed sets in X to closed sets in Y.

Let $C \subseteq X$ be a closed set, and since X is compact, so is C. Since f is continuous and C is compact, then $f(C)$ must also be compact since continuous images of compact sets are compact. Since Y is Hausdorff, and $f(C)$ is compact and a subset of Y, $f(C)$ must be closed in Y.

Therefore, since both C and $f(C)$ are closed for all $C \subseteq X$, f maps closed sets to closed sets.

> Question 9
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> Prove that if X is an ordered set in which every closed interval is compact, then X has the least upper bound property.

Proof: Let X be an ordered set and let each closed interval in X be compact. We must show that X has the least upper bound (LUB) property.

Suppose for the sake of contradiction that X does not have the LUB property. Then there exists a nonempty set $A \subseteq X$ that is bounded above but has no least upper bound. Let B be the set of all upper bounds of A. B is nonempty, and since A has no least upper bound, B has no smallest element.

Now pick any $a_0 \in A$ and $b_0 \in B$. Consider the closed interval $[a_0, b_0]$. For each $x \in [a_0, b_0]$:

- If $x \in B$, then since B has no smallest element, there exists $x' \in B$ with $x' < x$. Then $x \in (x', \infty)$, and we assign to x the open set (x', ∞).

- If $x \notin B$, then x is not an upper bound for A, so there exists $a \in A$ with $a > x$. Then $x \in (-\infty, a)$, and we assign to x the open set $(-\infty, a)$.
  This gives an open cover of $[a_0, b_0]$.

Since $[a_0, b_0]$ is compact, it has a finite subcover consisting of sets of the form $(-\infty, a_i)$ and (x'_j, ∞). Let a^* be the largest a_i. Then $a^* \in A$ must be covered by some (x', ∞), so $x' < a^*$. But $x' \in B$ is an upper bound for A and $a^* \in A$ exceeds it forming a contradiction.

Therefore, X has the LUB property.

Question 10

Let $\mathbb{R}_K$ denote $\mathbb{R}$ with the $K-$topology.

- Show that $[0, 1]$ is not compact as a subspace of $\mathbb{R}_K$.
- Show that $\mathbb{R}_K$ is connected.
- Show that $\mathbb{R}_K$ is not path connected.

Show that $[0, 1]$ is not compact as a subspace of $\mathbb{R}_K$
The K-topology has the standard open intervals (a, b) together with sets of the form $(a, b) \setminus K$ as a basis, where $K = \{1/n : n \in \mathbb{Z}^+\}$. Consider the open cover of $[0, 1]$, consisting of $(-1, 1) \setminus K$ together with the intervals $(1/n, 2)$ for each $n \in \mathbb{Z}^+$. Every point of $(0, 1]$ outside K is covered by $(-1, 1) \setminus K$, each point $1/n \in K$ is covered by $(1/(n+1), 2)$, and 0 is covered by $(-1, 1) \setminus K$. This cover has no finite subcover, since removing any $(1/n, 2)$ leaves $1/n$ uncovered. Thus $[0, 1]$ is not compact.

Show that $\mathbb{R}_K$ is connected.
Suppose for contradiction that $\mathbb{R}_K = U \cup V$ for some separation. Since the K-topology is finer than the standard topology, any separation of $\mathbb{R}_K$ would also be a separation of $\mathbb{R}$. But $\mathbb{R}$ is connected, so no separation exists. Thus $\mathbb{R}_K$ is connected.

Show that $\mathbb{R}_K$ is not path connected.
Suppose for the sake of contradiction that $f : [0, 1] \to \mathbb{R}_K$ is a continuous path with $f(0) = 0$ and $f(1) = 1$. Since $[0, 1]$ is compact and f is continuous, $f([0, 1])$ is a compact subspace of $\mathbb{R}_K$. But $f([0, 1])$ is a connected compact subset of $\mathbb{R}_K$ containing 0 and 1, so it must contain $[0, 1]$ (since $\mathbb{R}_K$ has the same connected subsets as $\mathbb{R}$). Thus $[0, 1] \subseteq f([0, 1])$, meaning $[0, 1]$ is a compact subspace of $\mathbb{R}_K$, contradicting *Part 1*.